BEE LAB PROJECT REPORT

Retail Sales Analytics: Excel & Power BI Dashboards

Student Name	Rahul Parihar
Roll Number	268
College	JECRC Foundation, Jaipur
Department	B.Tech First Year
Faculty Guide	Prof. Ram Singh
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1. Introduction & Background

Retail sales data contains rich insights about customer behaviour, product performance, and regional trends. This project uses Microsoft Excel and Microsoft Power BI to transform a 300-order, 3-year (2020–2022) retail dataset into two interactive Business Intelligence dashboards that help stakeholders make fast, data-driven decisions without requiring any programming knowledge.

2. Problem Statement

"Retail businesses generate large volumes of transactional data but often lack the tools to extract actionable insights. This project designs two interactive dashboards — one in Microsoft Excel and one in Microsoft Power BI — enabling stakeholders to monitor KPIs, compare category and regional performance, analyze profit margins, and forecast future sales trends without programming knowledge."

3. Objectives

- Build an interactive Excel Dashboard with Pivot Tables, PivotCharts, and Slicers.
- Develop a Power BI Dashboard with 16 DAX-powered KPI measures and a 3-month forecast.
- Perform Year-over-Year (YoY) and Month-over-Month (MoM) trend analysis.
- Identify top-performing and loss-making product sub-categories.
- Deliver a professional report and presentation for submission.

4. Methodology

Phase	Tool	Activity
1 — Data Collection	Kaggle / CSV	Downloaded retail sales dataset (300 rows, 2020-2022)
2 — Data Cleaning	Power Query	Fixed data types, standardized dates, removed nulls
3 — Data Modeling	Power BI DAX	Created DateTable, Star Schema, 16 DAX measures
4 — Dashboard Design	Excel + Power BI	Pivot Tables, PivotCharts, Slicers, Visuals, Forecast
5 — Analysis & Report	Both Tools	KPI tracking, YoY trends, findings, submission

5. Dataset Description

File: retail_sales_final.csv | Source: Kaggle Superstore (curated)

- Total Records: 300 orders across 3 years (2020, 2021, 2022)
- Categories: Technology ($2,50,980) | Furniture ($74,164) | Office Supplies ($9,251)
- Regions: West ($1,01,195) | South ($84,679) | East ($81,436) | Central ($67,086)
- Segments: Consumer (48%) | Corporate (30%) | Home Office (22%)
- 21 loss-making orders due to high discounts on Furniture

6. Data Dictionary

Column	Type	Description
Order ID	Text	Unique order identifier (CA-YYYY-XXXXXX)
Order Date	Date	Date order was placed (DD-MM-YYYY)
Ship Date	Date	Date order was shipped (DD-MM-YYYY)
Customer Name	Text	Full name of customer
Segment	Text	Consumer / Corporate / Home Office
Region	Text	East / West / Central / South
State	Text	US delivery state
Category	Text	Technology / Furniture / Office Supplies
Sub-Category	Text	Phones, Laptops, Chairs, Tables, Binders, etc.
Product Name	Text	Full product name
Sales	Decimal	Gross revenue ($)
Quantity	Integer	Units ordered
Discount	Decimal	Discount rate (0.10 = 10%)
Profit	Decimal	Net profit; negative = loss

7. Excel Dashboard — Store Sales Performance

File: Store_Sales_Dashboard_FINAL.xlsx | 6 sheets with charts and KPI cards

7.1 KPI Summary

KPI Metric	Value
Total Sales (2020–2022)	$3,34,395.01
Total Profit	$51,612.01
Profit Margin %	15.4%
Total Orders	300
Total Units Sold	699
2020 Sales	$81,018.48
2021 Sales	$1,31,145.77 (+61.9% YoY)
2022 Sales	$1,22,230.76 (-6.8% YoY)

7.2 Dashboard Sheets

Sheet Name	Contents
DASHBOARD	10 KPI cards, 3 PivotCharts, segment & year summary
SalesData	300-row formatted Excel Table
PT_Category	Sales & Profit by Category × Region + Bar Chart
PT_YearlyComparison	2020 vs 2021 vs 2022 monthly trend + Line Chart
PT_TopProducts	Top 10 products by profit + Horizontal Bar
PT_SubCategory	All sub-categories ranked by profit (losses in red)

7.3 Slicers & Interactivity

- Region Slicer — filters all charts simultaneously via Report Connections
- Category Slicer — isolates a single product category
- Segment Slicer — Consumer / Corporate / Home Office
- Year Slicer — compare any single year or multi-year view
- Date Timeline Slicer — drag to select custom date range

8. Power BI Dashboard — Retail Sales & Forecast BI

8.1 Data Model (Star Schema)

- SalesData (Fact Table) — 300 rows, 14 columns
- DateTable (Dimension) — created via DAX CALENDAR() function
- Relationship: DateTable[Date] —1:Many— SalesData[Order Date]
- DateTable marked as official Date Table for all time intelligence measures

8.2 Key DAX Measures

Measure	Formula (simplified)	Result
Total Sales	SUM(SalesData[Sales])	$3,34,395
Total Profit	SUM(SalesData[Profit])	$51,612
Profit Margin %	DIVIDE([Profit],[Sales],0)	15.4%
YTD Sales	TOTALYTD([Total Sales],DateTable[Date])	Cumulative
YoY Growth %	DIVIDE([Sales]-[Sales PY],[Sales PY],0)	+61.9% (2021)
MoM Growth %	DIVIDE([Sales]-[Sales PM],[Sales PM],0)	Monthly
Sales Previous Year	CALCULATE([Sales],SAMEPERIODLASTYEAR(...))	Benchmark
Loss Orders Count	CALCULATE([Orders],Profit<0)	21 orders

8.3 Report Pages

Page	Visuals
Overview	5 KPI Cards, Bar Chart, Line Chart, Donut Chart, Map Visual
Category Analysis	Matrix Table, Stacked Bar, Scatter Plot, KPI Card
Regional View	Filled Map, Column Chart, Summary Table, Treemap
Forecast	Line Chart + 3-month forecast (95% CI), MTD Sales, MoM Cards

8.4 Forecasting Setup

- Visual: Line Chart — X-Axis: DateTable[Date] (Month), Y-Axis: Total Sales
- Analytics Pane → Forecast → Add
- Forecast length: 3 months | Confidence Interval: 95% | Seasonality: Auto
- Output: Dashed forecast line with grey confidence band

9. Key Findings & Business Insights

- Technology is the top category at $2,50,980 revenue (75% of total) — Laptops & Phones lead.
- 2021 recorded the highest YoY growth at +61.9% — strong post-2020 recovery.
- 2022 saw a -6.8% decline — warrants investigation into pricing/discount strategy.
- West region leads all regions with $1,01,195 in total sales.
- Laptops ($19,678) and Phones ($19,621) are the most profitable sub-categories.
- Tables and Bookcases show losses due to heavy discounts (15–20%).
- Consumer segment contributes 48% of total revenue — the primary market.
- Forecast: Sales projected to stabilize and recover in early 2023.

10. Limitations

- Dataset is US-based; Indian retail patterns may differ significantly.
- Power BI forecasting uses exponential smoothing only — no external factor modeling.
- 300 records is sufficient for learning purposes; enterprise dashboards use millions of rows.
- No live database connection — CSV is static and not updated in real-time.
- No customer-level analytics (RFM, CLV) — out of scope for this project.

11. Conclusion

This project successfully demonstrates a complete Business Intelligence workflow — from raw retail CSV data to two fully interactive dashboards using Microsoft Excel and Power BI. The Excel dashboard provides Pivot Table-based KPI monitoring with Slicer interactivity, while the Power BI dashboard delivers advanced DAX analytics, a proper Star Schema data model, and 3-month sales forecasting with 95% confidence intervals. Both tools together address all project objectives of the BEE Lab and provide a strong foundation for enterprise-level retail analytics.

12. References

- [1] Microsoft DAX Reference — docs.microsoft.com/en-us/dax/
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